

Course Path

STAT 109 — Summer Session 1, 2026 (online, async)

This page shows the recommended path through the course during the five-week summer session from June 1 to July 2, 2026. You may work at your own pace within each module window, but the modules should be done in the order shown. Check [Canvas](#) to turn in work, take quizzes, see grades, and join discussions.

Each project module follows the same pattern: **Read 0** (project preview) → read → update your reference sheets → low-stakes quizzes (as many times as you want) → lab (Wednesday) → synthesis quiz (Thursday) → project milestone (Friday) → project (Sunday). Module 5 is shorter and ends on Thursday, July 2.

Reference sheets ([methods map](#), [R reference](#)) grow a little in every module 1–5. All readings also live on the [lectures](#) index if you want the full list.

Outcomes below are tagged **essential** (needed for the module project or core module goals) or **optional** (helpful enrichment or stretch).

The **Supplemental (optional)** column links extra **textbook chapters** (Vu & Harrington, free online) and **videos/tutorials** that match our Colab + `{tidyverse}` workflow — see [Resources](#). None of these are required for the module project unless a row says otherwise.

At a glance

Module	Dates	Project due
0 — Welcome	Open before Jun 1	—
1 — Testing a Claim	Jun 1–7	Sun Jun 7
2 — Estimating the Truth	Jun 8–14	Sun Jun 14
3 — Discovering Patterns	Jun 15–21	Sun Jun 21
4 — Designing Fair Comparisons	Jun 22–28	Sun Jun 28
5 — Looking Ahead	Jun 29–Jul 2	Thu Jul 2

Module 0 — Start here

Complete this before Module 1

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 0 Read 0	—	Essential: Welcome — five-step statistical process, three worlds, five portfolio projects, Colab/R notebook structure, and Module 0 checklist (labs, quizzes, research questions). ~15 min before the syllabus and labs.
Read	Syllabus , course path , resources	—	Essential: Know where readings, labs, and projects live vs what happens in Canvas. Essential: Understand grading, due-date pattern, and academic honesty / AI expectations (syllabus). Optional: Full policy details you will not need until later in the term.

Step	Materials	Supplemental (optional)	Learning outcomes
Quiz 0 (Start Here)	Canvas Module 0 — Quiz 0	—	Essential: Confirm Read 0 concepts: statistics definition, five-step process (ask → plan → collect → analyze → conclude), three worlds, Colab text/code/output, and basic R from the Module 0 lab. Open notes; unlimited attempts.
Quiz 0B (Question Ideas)	Canvas Module 0 — Quiz 0B	—	Essential: Submit 3 research questions for Step 1 practice; the third should compare two groups (Project 1 preview).
Reference sheets	Methods map , R reference	—	Essential: Set up two places to create your two references over the next 5 weeks. You will update these each module and can use them on assignments. Optional: Skim the methods map layout and R reference before Module 1 labs.
Lab (Wed)	Demo lab , Exercise lab	—	Essential: Run R in Google Colab (save a copy, run cells). Essential: Work with vectors, logical TRUE/FALSE, and simple simulation. Optional: Estimating probabilities by simulation (helps later probability intuition, not required for Project 1).
Canvas	Module 0 discussion	—	Essential: Introduce yourself and confirm you can reach assignments and discussions.
Supplemental pack	—	Colab tutorial · Posit: Getting Started with R · Textbook (free PDF) · Posit cheatsheets	Optional: Colab orientation and R basics before Module 1; textbook preface for context.

Module 1 — Testing a Claim (Jun 1–7)

Is this observation surprising?

Lab Wed Jun 3 · Synthesis Thu Jun 4 · Milestone Fri Jun 5 · [Project 1 — Testing a Claim](#) Sun Jun 7 · Discussion: something surprising about Lumberjacks (Canvas)

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 1 Read 0	—	Essential: Project preview — big question, sample ESP report (PDF) , what to notice in each section, link to Project 1 . ~5 min before Read 1.

Step	Materials	Supplemental (optional)	Learning outcomes
Read	Module 1 Read 1	—	Essential: Probability language through binomial pmf — sample spaces, rules, counting, and the binomial process.
Quizzes (Def & Notation)	Canvas Module 1 — Quiz A	—	Essential: Notation, sample space, addition rule, conditional probability, independence. Open notes; unlimited attempts.
Read more	Module 1 Read 2	—	Essential: Binomial test workflow and R — hypotheses, null plots, p-values.
Quizzes (Concept Check)	Canvas Module 1 — Quiz B	—	Essential: Counting, conditional applications, combinations, BRP identification. Open notes; unlimited attempts.
Reference sheets	Methods, R reference	—	Essential: Probability vocabulary (sample space through independence) on statistics sheet. Essential: One-proportion / binomial test row on methods map. Essential: R patterns for <code>dbinom</code> , <code>pbinom</code> , <code>geom_col</code> , <code>geom_vline</code> .
Lab (Wed)	Demo lab, Conditional demo, Exercise lab, Project 1 milestone	—	Essential (demos): simulation review, data structures, conditional simulation, binomial R tools, null plot. Essential (exercise): combined practice notebook (Parts I–III). Essential (milestone): Project 1 precheck. Update R reference .
Synthesis quiz (Thu)	Canvas Module 1 — Synthesis	—	Essential: Full binomial test workflow, Module 1 R practice, interpretation items. After labs. Two attempts; open notes.
Milestone (Fri)	Project 1 milestone	—	Essential: Phase A — question, p , H_0/H_a , null graph before data collection.
Project (Sun)	Project 1 — Testing a Claim	—	See project outcomes table below.
Supplemental pack	—	Textbook Ch 2 — Probability · Ch 3.2 — Binomial · Khan: binomial distribution · Khan: visualizing binomial	Optional: Alternate probability explanations; course labs use <code>{ggplot2}</code> for null plots, not the textbook’s base-R labs.

Project 1 — Testing a Claim

Outcome	Tag
Choose a real binomial process ($n = 25$) that is not a class example; defend all four BRP conditions in your own context	Essential
State H_0 and H_a for population proportion p with a numeric null value	Essential

Outcome	Tag
Report x , n , $\hat{p}$; descriptive bar/pie chart in R	Essential
Plot null binomial distribution with <code>dbinom</code> + <code>geom_col</code> ; mark observed x	Essential
Compute p-value with <code>pbinom</code> matching alternative direction; interpret in one sentence	Essential
One-page report with Introduction / Data Collection / Data Analysis / Conclusion	Essential
Use the Read 2 ESP worksheet as a model without copying ESP as your study	Essential

Module 2 — Estimating the Truth (Jun 8–14)

Building high-quality estimates from samples.

Lab Wed Jun 10 · Synthesis Thu Jun 11 · Milestone Fri Jun 12 · **Project 2 — Estimating the Truth** Sun Jun 14 · Discussion: a plant you appreciate (Canvas)

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 2 Read 0	—	Essential: Project preview — finite population, SRS, 95% CI focus; tour sample report (PDF) . ~5 min before Read 1.
Read	Module 2 Read 1	—	Essential: Binomial $\mu = np$, $\sigma = \sqrt{np(1-p)}$; normal approximation & Empirical Rule; sampling distribution & 95% CI (proportion and mean); sampling frame, bias, precision, SRS and design types.
Quizzes (Def & Notation)	Canvas Module 2 — Quiz A	—	Essential: Vocabulary for estimation and sampling (frame, SRS, bias, CI language). Open notes; unlimited attempts.
Read more	Module 2 Read 2	Textbook Ch 4.2 — Confidence intervals	Essential: z-scores & standard normal; methods-map habit; mean/median, histogram, dot plot in tidyverse. Optional: <code>prop.test()</code> / one-proportion z-test; one-sample <code>t.test()</code> (Project 2 uses CI only).
Quizzes (Concept Check)	Canvas Module 2 — Quiz B	—	Essential: z-scores, toolbox organization, one-variable summaries. Open notes; unlimited attempts.
Reference sheets	Methods, R reference	—	Essential: One-mean and one-proportion CI rows; FPC note for finite populations. Essential: z-score row before toolbox section.

Step	Materials	Supplemental (optional)	Learning outcomes
Lab (Wed)	Module 2 demo , Exercise lab	—	Essential: Choose among CI situations (mean vs proportion; finite N ; FPC). Essential: Check conditions and compute 95% CI in R.
Synthesis quiz (Thu)	Canvas Module 2 — Synthesis	—	Essential: CI computation and interpretation; SRS vs convenience; FPC decision; exercise-lab R patterns; tidyverse summaries. After labs.
Milestone (Fri)	Canvas Module 2	—	Essential: Named finite population (N), variable type, planned SRS method, and which CI type you will use.
Project (Sun)	Project 2 — Estimating the Truth	—	See project outcomes table below.
Supplemental pack	—	Textbook Ch 3.3 — Normal · Ch 4.1–4.2 — Variability & CIs · Khan: confidence intervals · Khan: CI simulation	Optional: CI intuition and normal-model background; Project 2 rubric is CI-focused.

Project 2 — Estimating the Truth

Outcome	Tag
Define a listable finite population (N) and one variable (binary $\rightarrow$ proportion, or numerical $\rightarrow$ mean)	Essential
Draw an SRS of size n from $\{1, \dots, N\}$ without replacement; collect data	Essential
Compute point estimate ($\hat{p}$ or $\bar{x}$)	Essential
Choose correct 95% CI (mean t -interval or proportion interval; apply FPC when N is finite)	Essential
State and check interval conditions in context	Essential
Interpret CI with the “I am 95% confident that...” sentence	Essential
Submit one-page summary + runnable R notebook (mean or proportion template)	Essential
Sample size calculation / professional power analysis	Optional
Hypothesis test about the parameter (not required by rubric)	Optional

Module 3 — Discovering Patterns (Jun 15–21)

Describing a dataset with graphs and summaries.

Lab Wed Jun 17 · Synthesis Thu Jun 18 · Milestone Fri Jun 19 · [Project 3 — Discovering Patterns](#) Sun Jun 21 · Discussion: a place from home (Canvas)

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 3 Read 0	—	Essential: Project preview — EDA goals, variable-type table, sample Project 3 tour. ~5 min before Read 1.
Read	Module 3 Read 1	—	Essential: Compare numerical outcome across two groups with spread (range, IQR, SD), boxplots, and <code>group_by()</code> + <code>summarize()</code> . Optional: Welch two-sample <i>t</i> -test (Project 3 is descriptive only).
Quizzes (Def & Notation)	Canvas Module 3 — Quiz A	—	Essential: Tidyverse pipeline verbs (<code>filter</code> , <code>select</code> , <code>mutate</code> , <code>count</code> , <code>group_by</code> , <code>summarize</code>). Essential: Variable types (numerical vs categorical). Essential: Spread vocabulary (range, quartiles, IQR, SD, five-number summary). Open notes; unlimited attempts.
Read more	Module 3 Read 2	—	Essential: Scatterplots for two numerical variables; form, direction, strength. Essential: Two-way tables (joint/row/column proportions) and stacked bar charts.
Quizzes (Concept Check)	Canvas Module 3 — Quiz B	—	Essential: Choose the right graph/summary by variable type. Essential: Association description. Essential: Two-way table reading. Open notes; unlimited attempts.
Reference sheets	Methods, R reference	—	Essential: Describe rows for one categorical, one numerical, two categorical, two numerical, and numerical-by-group. Essential: <code>ggplot</code> patterns for histogram, boxplot, bar, scatter, stacked bars. Optional: Inferential rows for two-group tests (not required for Project 3).
Lab (Wed)	Module 3 demo , Exercise lab	—	Essential: Five EDA situations: univariate numerical/categorical; boxplots by group; scatterplot; two-way table + stacked bars. Essential: <code>group_by</code> + <code>summarize</code> ; <code>count</code> + <code>pivot_wider</code> .
Synthesis quiz (Thu)	Canvas Module 3 — Synthesis	—	Essential: Five-number summary; compare groups from boxplots; scatterplot description; two-way proportions; <code>ggplot2</code> calls. After labs. Two attempts; open notes.
Milestone (Fri)	Canvas Module 3	—	Essential: Chosen dataset link, four variables with types, and plan for two figures + four numeric facts.

Step	Materials	Supplemental (optional)	Learning outcomes
Project (Sun)	Project 3 — Discovering Patterns	—	See project outcomes table below.
Supplemental pack	—	Textbook Ch 1 — Intro to data (§1.4–1.7) · R4DS: Data visualization · Posit: ggplot2 video	Optional: EDA and <code>{ggplot2}</code> depth; use course Colab labs for <code>{tidyverse}</code> practice (not the textbook’s base-R lab sets).

Project 3 — Discovering Patterns

Outcome	Tag
Choose a dataset (TidyTuesday or other) with 4 variables and 10 rows; document source	Essential
Classify four variables as numerical or categorical	Essential
Univariate summaries and plots for each of the four variables	Essential
Bivariate exploration for variable pairs (tables/plots appropriate to types)	Essential
One-page PDF: context, two best graphics, 4 labeled numeric summaries, four findings	Essential
Notebook runs top-to-bottom with all code used (4 univariate + 6 bivariate plots in notebook; only 2 plots on one-pager)	Essential
Part A short answers in Canvas	Essential
Inferential tests or population generalization beyond the dataset	Optional (explicitly out of scope — descriptive statistics only)
Your own research question beyond exploratory findings	Optional

Module 4 — Designing Fair Comparisons (Jun 22–28)

Comparing two groups responsibly.

Lab Wed Jun 24 · Synthesis Thu Jun 25 · Milestone Fri Jun 26 · **Project 4 — Designing Fair Comparisons** Sun Jun 28 · Discussion: a place in Humboldt (Canvas)

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 4 Read 0	—	Essential: Project 4 preview; what makes a two-group comparison fair; vocabulary overview.
Read 1	Module 4 Read 1	—	Essential: EV vs RV; observational studies vs experiments; confounding; DAG with one confounder; “by chance” vs causal language. Essential: RCT definition; what random assignment buys. Optional: Adjustment sets / backdoor criterion beyond one confounder.

Step	Materials	Supplemental (optional)	Learning outcomes
Reference sheets	Methods, R reference	—	Essential: Two-group inferential rows (Welch t , two-proportion / χ^2 as appropriate). Essential: Study-design vocabulary. Essential: R pipeline: <code>group_by</code> , plots by group, <code>t.test</code> / <code>prop.test</code> / <code>chisq.test</code> .
Quiz A (Def & Notation)	Canvas Module 4 — Quiz A	—	Essential: RCT vs observational; confounder; DAG; causal vs association language.
Read 2	Module 4 Read 2	—	Essential: Pick two-group test from methods table; Welch t-test; two-proportion z-test; conditions; connect random assignment to reduced confounding.
Quiz B (Concept Check)	Canvas Module 4 — Quiz B	—	Essential: Design classification; methods-map matching; read two-group R output.
Demo lab	Module 4 demo lab	—	Essential: <code>set.seed()</code> ; numerical outcome $\rightarrow$ Welch t-test; binary outcome $\rightarrow$ two-proportion z-test; validate under null; RCT vs observational confounding demo.
Exercise lab (submit)	Module 4 exercise lab	—	Essential: Four problems — same structure as demo. Connect simulation plan to Project 4 design.
Synthesis quiz	Canvas Module 4 — Synthesis	—	Essential: Design + methods table integration; interpret two-group CIs; causal vs association language; simulation R workflow; error types in context.
Milestone	Canvas Module 4	—	Essential: Assigned research question; draft answers to Part 1 design questions; plan for simulated data structure.
Project (Sun)	Project 4 — Designing Fair Comparisons	—	See project outcomes table below.
Supplemental pack	—	Textbook Ch 5.3 — Two-sample tests · Ch 8.1–8.2 — Proportions · R4DS: Data transformation	Optional: Two-group inference concepts; course labs use <code>group_by()</code> , <code>{ggplot2}</code> , and Colab.

Project 4 — Designing Fair Comparisons

Outcome	Tag
Answer ten study-design questions (EV/RV, operational definitions, hypotheses, RCT vs observational, confounder + DAG, test name from methods table)	Essential
Simulate reproducible dataset (<code>set.seed</code>) with binary EV, response, and confounder when needed	Essential

Outcome	Tag
Descriptive summaries and plots by group (numerical → boxplot/summaries; binary → stacked bars / proportions)	Essential
Run appropriate two-group test / CI; report p-value and interval; discuss “by chance” explanation	Essential
One-page summary written for the classmate who asked the question	Essential
Runnable notebook with simulation + analysis sections	Essential
Real-world data collection for the assigned question	Optional (project uses simulation)
Adjust for confounding in the inferential model	Optional (confounder included in design/DAG and plots; test does not remove confounding)

Module 5 — Looking Ahead (Jun 29–Jul 2)

Using data to make predictions.

This module is four days. Lab Wed Jul 1 · **Project 5 — Looking Ahead**, synthesis quiz, milestone, and portfolio package due Thu Jul 2 · Discussion: predicting a future observation (Canvas)

Step	Materials	Supplemental (optional)	Learning outcomes
Read 0	Module 5 Read 0	—	Essential: Project preview — nesting dolls or stones; four-day schedule; what to expect in Read 1–2 and the lab. ~5 min before Read 1.
Read	Module 5 Read 1	Regression line applet	Essential: Scatterplot description (form, direction, strength); Pearson r ; fit <code>lm(y ~ x)</code> ; interpret slope and intercept; point prediction; residuals; MSE and R^2 on test set.
Quizzes (Def & Notation)	Canvas Module 5 — Quiz A	—	Essential: Scatterplot vocabulary, regression notation, slope/intercept interpretation, residual definition, R^2 . Open notes; unlimited attempts.
Read more	Module 5 Read 2	Penn State: prediction intervals	Essential: Residual normality condition; 95% CI for slope with <code>confint()</code> ; CI for mean response vs prediction interval with <code>predict(..., interval=)</code> ; why PI is wider. Optional: Textbook Ch 6.5 on interval estimates with regression.
Quizzes (Concept Check)	Canvas Module 5 — Quiz B	—	Essential: Residual conditions, CI for slope, mean-response CI vs prediction interval. Open notes; unlimited attempts.
Reference sheets	Methods, R reference	—	Essential: Two-numerical describe + regression / prediction rows. Essential: <code>lm</code> , <code>predict</code> , scatterplot with fitted line on R sheet.

Step	Materials	Supplemental (optional)	Learning outcomes
Lab (Wed)	Demo lab, Exercise lab	—	Essential (demo): Scatterplot + <code>cor</code> ; fit <code>lm</code> ; point prediction; <code>confint</code> for slope; <code>predict</code> with CI and PI. Essential (exercise): Apply same five-step workflow to your team's Project 5 data.
Synthesis quiz (Thu)	Canvas Module 5 — Synthesis	—	Essential: Read scatterplots; interpret r , slope, intercept; prediction vs confidence interval; read <code>predict</code> output; portfolio method review. After labs. Two attempts; open notes.
Milestone (Thu)	Canvas Module 5	—	Essential: Team, option (nesting dolls or stones), variables defined, data collection underway.
Project (Thu)	Project 5 — Looking Ahead	—	See project outcomes table below.
Supplemental pack	—	Textbook Ch 6 — Simple linear regression · Khan: regression line example · Penn State: prediction intervals	Optional: Regression concepts and PI vs CI; Project 5 requires SLR only.

Project 5 — Looking Ahead

Outcome	Tag
Form a team (1–4); choose Option A (nesting dolls) or B (stone container)	Essential
Define measurement rules and collect team dataset (CSV / shared sheet)	Essential
Fit at least one simple linear model <code>lm(response ~ predictor)</code>	Essential
Report prediction equation, plugged prediction for target case, 95% prediction interval	Essential
One-page PDF: intro, methods, results, conclusion, team contributions	Essential
Runnable team <code>.ipynb</code> from data load through model and prediction	Essential
Compare multiple models / predictors and choose best	Optional (encouraged but one SLR minimum)
Multiple predictors / logistic regression for binary outcomes	Optional (beyond Project 5 scope)
Portfolio package: both reference sheets + all five projects polished for submission	Essential

Notes

- **Supplemental (optional)** links point to the [Vu & Harrington textbook](#) and Colab/{tidyverse}-friendly videos — not required for the module project. The textbook's companion R labs use base R; prefer this course's Colab notebooks and [Resources](#).
- Modules unlock in order in Canvas ($0 \rightarrow 1 \rightarrow \dots \rightarrow 5$).
- Quizzes are open notes: use your reference sheets and the methods map.
- Definitions & Notation and Concept Check: unlimited attempts. Synthesis: two attempts (highest score counts).

- The milestone is an ungraded checkpoint — it helps you and me catch problems before the project is due.
- Modules 1–4: lab Wednesday, synthesis Thursday, milestone Friday, project Sunday.
- Module 5: lab Wednesday July 1; synthesis, milestone, project, and portfolio package due Thursday July 2.
- Quiz source files (**quizzes/quiz-*.qmd**) are practice banks; Canvas quizzes may use a subset. Items marked **optional** above may still appear — they are not required for the module project.
- If dates shift when the term opens, Canvas is the official calendar; this page will be updated to match.